

SAFETY DATA SHEET

Preparation Date: August 28, 2017

Revision Date: October 9, 2017

Revision Number: 1

1. IDENTIFICATION

PRODUCT IDENTIFIER

Product Name: MULTIFILM

EMERGENCY TELEPHONE NUMBER

Canutec (613) 996-6666

OTHER MEANS OF IDENTIFICATION

Chemical Name: Not Applicable

Synonyms: Not Applicable

RECOMMENDED USE OF CHEMICAL

Closed Loop – Corrosion Inhibitor

RESTRICTIONS ON USE

Not Applicable

SUPPLIER IDENTIFICATION

Keytech Water Management

33 McIntyre Drive, Kitchener, ON N2R 1E4

1-800-265-2772

2. HAZARDS IDENTIFICATION

Classification:

Physical Hazards: Not classified

Health Hazards: Acute Toxicity – 4 – Oral
Skin Corrosion – 1
Serious Eye Damage – 1
Reproductive Toxicity – 2

Environmental Hazards: Hazardous to the Aquatic Environment

Label Elements



Signal Words: DANGER



Hazard Statements

Physical Hazards: None identified

Health Hazards: H302: Harmful if swallowed.
H314: Causes severe skin burns and eye damage.
H318: Causes serious eye damage.
H361: Suspected of damaging fertility or the unborn child.

Environmental Hazards: H413: May cause long lasting harmful effects to aquatic life.

Precautionary Statements

Prevention: P201: Obtain special instructions before use.
P202: Do not handle until all safety precautions have been read and understood.
P260: Do not breathe mist/vapours/spray.
P264: Wash face, hands and any exposed skin thoroughly after handling.
P270: Do not eat, drink or smoke when using this product.
P273: Avoid release to the environment.
P280: Wear protective gloves/protective clothing/eye protection/face protection.

Response: P301+P330+P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303+P361+P353: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.
P304+P340: IF INHALED: Remove person to fresh air and keep at rest in a position comfortable for breathing.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses if present and easy to do. Continue rinsing.
P310: Immediately call a POISON CENTER or doctor/physician.
P308+P313: If exposed or concerned: Get medical advice or attention.
P321: Specific treatment (see first aid instructions on the SDS).
P363: Wash contaminated clothing before reuse.

Storage: P405: Store locked up.

Disposal: P501: Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified (HNOC): None identified

3. COMPOSITION / INFORMATION ON INGREDIENTS

Hazardous Ingredients	CAS Number	% by Weight
Sodium nitrite	7632-00-0	9.5-10.5
Sodium metaborate	7775-19-1	2.7-3.0
Sodium hydroxide	1310-73-2	0.4-1.0
Sodium tolytriazole	64665-57-2	<0.1

Any components not listed are not hazardous according to WHMIS classification.

4. FIRST AID MEASURES

General Advice: Show this safety data sheet to the doctor in attendance.

Protection of First-aiders: Use personal protective equipment as per Section 8. Avoid contact with skin, eyes, clothing.

Eye Contact: Flush with water for a minimum of 15 minutes, holding eyelids open. Remove contact lenses if present and easy to do. Continue rinsing. If irritation persists, repeat flushing. Obtain medical attention.

Skin Contact: Immediately remove contaminated clothing and shoes and wash before reuse. Rinse and wash skin thoroughly with soap and water for at least 15 minutes. If irritation persists after initial washing, obtain medical attention.

Inhalation: Move to fresh air and keep at rest in a position comfortable for breathing. Give artificial respiration ONLY if breathing has stopped. If breathing is difficult, trained personnel should administer oxygen. Obtain medical attention if the person feels unwell after exposure.

Ingestion: Do not induce vomiting. Give 1 to 2 glasses of water to dilute the material if the victim is alert and not convulsing. Never give anything by mouth to an unconscious person. If spontaneous vomiting occurs, have victim lean forward with head down to avoid breathing in of vomitus. Rinse mouth and administer more water. Obtain medical attention if the person feels unwell after exposure.

Most important symptoms and effects, both acute and delayed: Irritation, nausea, headache, shortness of breath.

Notes to Physician: Treat symptomatically.

5. FIRE – FIGHTING MEASURES

Suitable Extinguishing Media: Use water spray. Do not use any extinguishing agent other than water. Use of heavy stream of water may spread the fire.

Hazardous Combustion Products: Oxides of carbon, sulphur, sodium, nitrogen, phosphorus.



Protective Equipment and Precautions for Firefighters: Wear NIOSH approved self-contained breathing apparatus and full protective clothing in case of fire. Remove storage containers from fire zone if possible. Use water to cool containers. Control run-off water by containing and keeping it out of sewers and watercourses.

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions: Avoid all contact with skin, eyes, or clothing. Avoid breathing (vapor, mist, spray).

Protective Equipment: Wear protective equipment/clothing as per Section 8.

Emergency Procedures: Restrict access to area until the spill has been cleaned up.

Environmental Precautions: Prevent from reaching drains, sewer or waterway. Product should not be released into environment.

Methods for Cleaning up: Block potential routes to water systems. Pump into appropriate containers or absorb with an inert material such as sand. Rinse residue with water. Dispose of waste and residues in accordance with local authority requirements.

7. HANDLING AND STORAGE

Precautions for Safe Handling: Wear protective equipment/clothing as per Section 8. Avoid contact with eyes, skin and clothing. Do not swallow. Avoid breathing mist/vapours/spray. Wash face, hands and any exposed skin thoroughly after handling. Do not eat, drink or smoke when using this product.

Conditions for Safe Storage: Store in a cool, dry, well ventilated location away from incompatible materials. Protect from freezing. Keep in original container. Keep container tightly closed.

Incompatible Materials: Oxidizing agents, reducing agents, acids, ammonium salts, amines.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure Guidelines

Chemical Name	ACGIH TLV	OSHA PEL	NIOSH IDLH
Sodium hydroxide	2 mg/m ³ ceiling	2 mg/m ³	10 mg/m ³

Appropriate Engineering Controls

Engineering Controls: Ensure eye wash stations and safety showers (or deluge hose) are proximal to the workstation location. Use process enclosures, local exhaust ventilation, or other engineering controls to keep airborne levels below recommended exposure limits.

Personal Protective Equipment

Eye and Face Protection: Wear chemical goggles. Use face shield if splashing is anticipated.

Skin and Body Protection: Wear chemical resistant gloves. Rubber, neoprene or PVC should be impervious under conditions for use. Skin contact should be prevented through the use of suitable protective clothing selected for conditions of use and exposure potential.

Respiratory Protection: A NIOSH-approved respirator with organic vapor filter should be used in environments without proper ventilation, where exposure guideline levels may be exceeded or if misting may occur.

General Hygiene Considerations: Handle in accordance with good industrial hygiene and safety practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Clear, yellow to yellow-green

Physical State: Liquid

Odour: N/D

Odour Threshold: N/D

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
pH	12.2-13.2	
Melting / Freezing Point	N/D	
Boiling Point / Range	N/D	
Flash Point	None	not flammable
Evaporation Rate	N/AP	
Flammability (solid, gas)	N/AP	
Flammability Limits in Air	N/AP	
Upper flammability limit	N/AP	
Lower flammability limit	N/AP	
Vapour Pressure	N/AP	
Vapour Density	N/AP	
Specific Gravity	1.085-1.105	
Water Solubility	Complete	
Solubility in other solvents	N/D	
Partition coefficient: n-octanol/water	N/D	
Auto-Ignition Temp (°C)	N/D	
Decomposition Temperature	N/D	
Viscosity, Kinematic	N/D	
Viscosity, Dynamic	N/D	

10. STABILITY AND REACTIVITY

Reactivity: Nonreactive under normal conditions.



Chemical Stability: Stable under normal conditions.

Possible Hazardous Reactions: None under normal processing.

Conditions to Avoid: Incompatible materials, excessive heat.

Incompatible Materials: Oxidizing agents, reducing agents, acids, ammonium salts, amines.

Hazardous Decomposition Products: Oxides of carbon, sulphur, sodium, nitrogen, phosphorus.

11. TOXICOLOGICAL INFORMATION

Component Information

Chemical Name	LD50 Oral	LD50 Dermal	LC50 Inhalation
Sodium nitrite	181 mg/kg rat	N/AV	5.5 mg/L/4h rat
Sodium metaborate	3251 mg/kg rat	N/AV	N/AV
Sodium hydroxide	500 mg/kg rabbit	N/AV	N/AV
Sodium tolyltriazole	640 mg/kg rat	N/AV	N/AV

Symptoms of Acute or Chronic Exposure and Delayed Effects

Ingestion: Harmful if swallowed. Ingestion may cause methemoglobinemia. Initial manifestation of methemoglobinemia is cyanosis, characterized by navy lips, tongue and mucous membranes, with skin color being slate grey. Further manifestation is characterized by headache, weakness, dyspnea, dizziness, stupor, respiratory distress and death due to anoxia. Signs and symptoms of nitrite poisoning include methemoglobinemia, nausea, dizziness, increased heart rate, hypotension, fainting and, possibly shock.

Inhalation: Harmful if inhaled. May aggravate existing asthma and inflammatory or fibrotic pulmonary disease.

Eye Damage / Irritation: Causes serious eye damage. Immediate effects may include burns of the mouth, oesophagus and stomach, with severe pain, vomiting, diarrhea. Damage may appear days after exposure. Severity is generally determined by the concentration of the solution and the duration of exposure.

Skin Corrosion / Irritation: Causes severe skin burns or ulcers. Severity is generally determined by the concentration of the solution and the duration of exposure. May aggravate existing dermatitis.

Sensitization: None identified.

Reproductive Effects: No data on the product itself. However, animal feeding studies with sodium tetraborate and boric acid in rat, mouse and dog, at high doses, have demonstrated effects on fertility and testes. Studies with the chemically related boric acid in rat, mouse and rabbit, at high doses, demonstrate developmental effects on the fetus including fetal weight loss and minor skeletal variations. The lowest NOAEL is 9.6 mg B/kg in rats, based on developmental effects. The doses administered were many times in excess of those to which humans would normally be exposed.



Mutagenic Effects: None identified.

Specific Target Organ Toxicity – Single Exposure: None identified.

Specific Target Organ Toxicity – Repeated Exposure: None identified.

Aspiration Hazard: None identified.

Carcinogenicity: Not listed as a carcinogen. Not expected to have mutagenic or reproductively toxic effects on humans.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Chemical Name	Freshwater Fish Species Data	Aquatic Invertebrate Data	Freshwater Algae Data	Microorganism Data
Sodium nitrite	LC50 0.092-0.13 mg/L 96 h Rainbow trout	N/AV	N/AV	N/AV
Sodium metaborate	LC50 65 mg B/L 24 d Rainbow trout	LC50 242 mg B /L 48 h Daphnia magna	EC10 24 mg B /L 96 h S.subspicatus	N/AV
Sodium hydroxide	LC50 45.4 mg/L 96 h Rainbow trout	N/AV	N/AV	N/AV
Sodium tolyltriazole	LC50 25 mg/L 96 h S. gairdneri	LC50 280 mg/L Daphnia magna	EC50 26.2 mg/L 72 h S.capricornutum	N/AV

Persistence and Degradability: Not established.

Bioaccumulation Potential: Not established.

Mobility in Soil: Not established.

Other Adverse Effects: None identified.

13. DISPOSAL CONSIDERATIONS

Waste Disposal Method: Dispose of waste and residues in accordance with local authority requirements.

Contaminated Packaging: Dispose of empty containers as unused product. Product or containers must not be disposed with household garbage.



14. TRANSPORT INFORMATION

UN #: UN 1760
UN Proper Shipping Name: CORROSIVE LIQUIDS, N.O.S. (sodium hydroxide)
Transport Hazard Class: 8
Packing Group: II

Special Precautions: No additional precautions required.

Environmental Hazards: May cause long lasting harmful effects to aquatic life.

15. REGULATORY INFORMATION

No additional regulatory information applicable.

16. OTHER INFORMATION

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SDSs available at: sds@keytech.ca, keytechservice.com

CAS: Chemical Abstracts Service (division of the American Chemical Society)

ACGIH: American Conference of Governmental industrial Hygienists

TLV: Threshold Limit Value – the reasonable level to which a worker may be repeated exposed over a working lifetime without adverse health effects

TWA: Time Weighted Average – a TLV based on an 8 hr work day or 40 hr work week

STEL: Short Term Exposure Limit – a TLV based on a 15 minute average

Ceiling – a TLV that should not be exceeded during any part of the work experience

OSHA: Occupational Safety and Health Administration (USA)

PEL: Permissible Exposure Limit – a TLV established by OSHA for exposure of an employee to a substance / physical agent

NIOSH: National Institute for Occupation Health and Safety

IDLH: Immediately Dangerous to Life or Health

CFR: Code of Federal Regulations (USA)

LD50: Lethal Dose – the amount of a material, given all at once, causing the death of 50% of a group of test animals.

LC50: Lethal Concentration – the concentration of a material in the air or water which causes the death of 50% of a group of test animals in a set period of time (usually 4 hours).

IC50: Inhibitory Concentration – measures the effectiveness of a substance in inhibiting a specific biological function.



N/AP = Not Applicable

N/AV = Not Available

N/D = Not Determined

Disclaimer

The information provided on this SDS is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guide for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered as a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.